



Dynamics and Control (UESTC 3001)



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- Lecture 1: Root Locus Analysis
- Lecture 2: Root Locus II and Nyquist Plots
- Lecture 3: Bode Plots
- Lecture 4: Bode Plots II
- Lecture 5: Stability in Frequency Domain
- Lecture 6: Stability cont. and Stability Examples
- Lecture 7: Compensators
- Lecture 8: Tutorials and Test Exercises

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Intended Learning Objectives

At the end of this lecture, you will work through:

- Stability cont.
- Non-unity feedback systems
- Examples on stability regions
- Examples on stability margins

Stability Margins

- The point of intersection of the polar plot with the negative real axis can be determined by setting the imaginary part of $G(j\omega)H(j\omega)$ equal to zero

$$G(j\omega)H(j\omega) = \mathbf{u} + \mathbf{jv} = \frac{-K(T_1 + T_2) - \mathbf{j}K\left(\frac{1}{\omega}\right)(1 - \omega^2 T_1 T_2)}{1 + \omega^2(T_1^2 + T_2^2) + \omega^4 T_1^2 T_2^2}$$

If we set the frequency at the point of intersection to be ω_2 , we have:

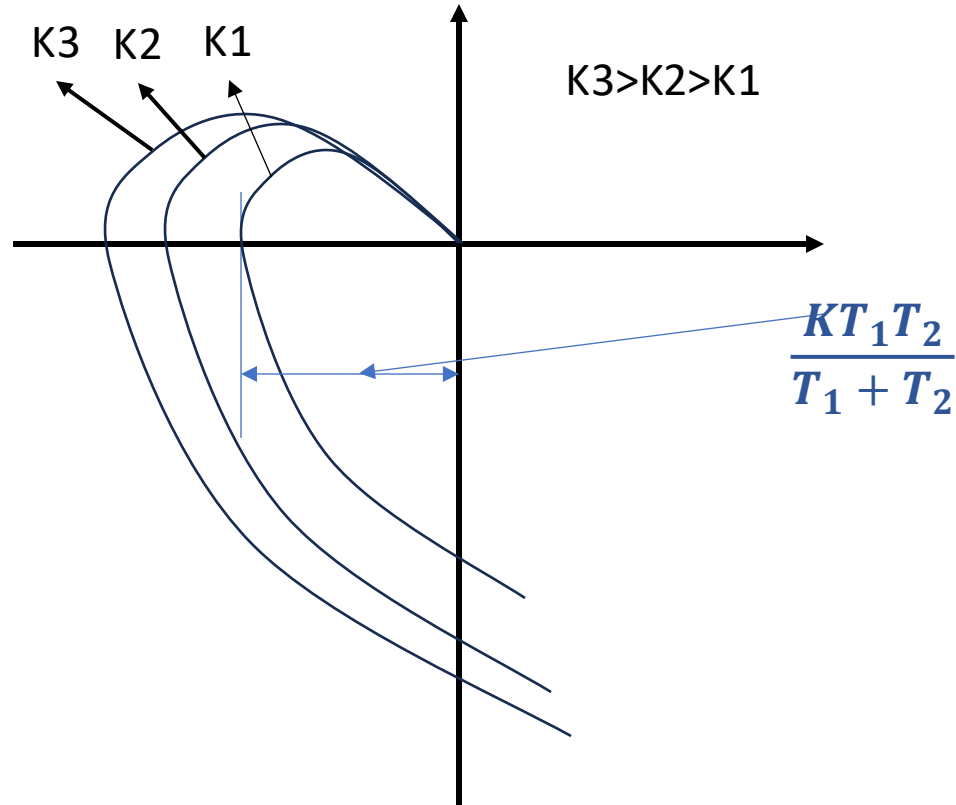
$$\mathbf{v} = \frac{-K\left(\frac{1}{\omega_2}\right)(1 - \omega_2^2 T_1 T_2)}{1 + \omega_2^2(T_1^2 + T_2^2) + \omega_2^4 T_1^2 T_2^2} = 0$$

This gives $\omega_2 = \frac{1}{\sqrt{T_1 T_2}}$

Stability Margins

- The magnitude of the real part at frequency ω_2 is given by:

$$u = \frac{-K(T_1 + T_2)}{1 + \omega^2(T_1^2 + T_2^2) + \omega^4 T_1^2 T_2^2} = -\frac{KT_1 T_2}{T_1 + T_2}$$



For the system to be stable,

$$\frac{KT_1 T_2}{T_1 + T_2} < 1 \text{ or } K < \frac{T_1 + T_2}{T_1 T_2}$$

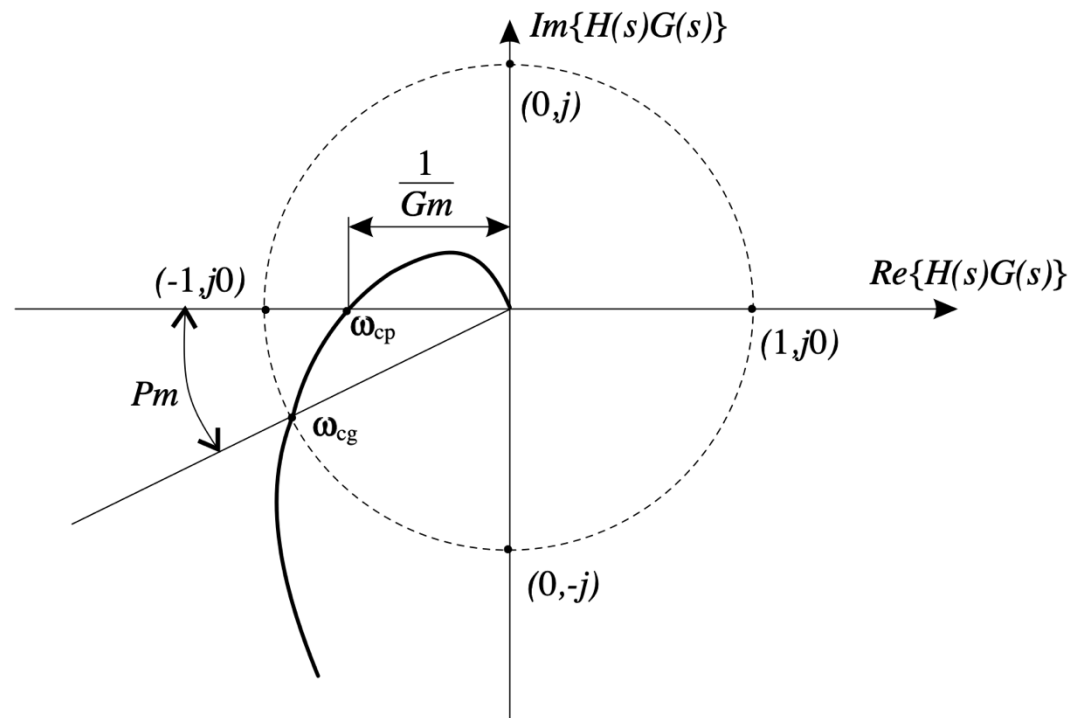
Recall: Margins

- The gain margin is defined as the factor by which you would have to increase the gain of a system to make it unstable.
- It is computed by determining the magnitude , λ , when ,
 $arg\{G(j\omega)\} = -180^{\circ}$, and defined as follows: $\text{Gain Margin} = \frac{1}{\lambda}$
- The phase margin, ϕ , is that amount of additional phase lag that could be added at the gain crossover frequency required to bring the system to the verge of closed loop instability.
- It is computed by determining the phase , θ , when $|G(j\omega)| = 1$ and adding this phase to 180 deg. *Phase Margin*, $\phi = 180^{\circ} + \theta$

Nyquist Margins

$$Pm = 180^\circ + \arg\{G(j\omega_{gc}), H(j\omega_{gc})\}$$

$$Gm[dB] = 20 \log \frac{1}{|G(j\omega_{pc})H(j\omega_{pc})|} [dB]$$



Correlation between PM and Damping Factor ζ

- Consider a unity feedback second order system with an open-loop transfer function

$$G(s)H(s) = \frac{K}{s(\tau s + 1)} = \frac{\omega_n^2}{s(s + 2\zeta\omega_n)}$$

where $\omega_n = \sqrt{\frac{K}{\tau}}$ and $2\zeta\omega_n = \frac{1}{\tau}$

Replacing s by $j\omega$ for obtaining a polar plot, we have

$$G(j\omega)H(j\omega) = \frac{\omega_n^2}{j\omega(j\omega + 2\zeta\omega_n)}$$

Finding the phase margin can show that for $\zeta \leq 0.7$, $\zeta \approx 0.01\phi$, where ϕ is the phase margin.



Non-unity feedback systems

Non-unity feedback systems

- For a non-unity feedback system, the closed loop transfer function is given by

$$\begin{aligned} T(j\omega) &= \frac{C(j\omega)}{R(j\omega)} = \frac{G(j\omega)}{1 + G(j\omega)H(j\omega)} = \frac{1}{H(j\omega)} \left[\frac{G(j\omega)H(j\omega)}{1 + G(j\omega)H(j\omega)} \right] \\ &= \frac{1}{H(j\omega)} \left[\frac{G_o(j\omega)}{1 + G_o(j\omega)} \right] = \frac{1}{H(j\omega)} T_o(j\omega) \end{aligned}$$

Where $G_o(j\omega) = G(j\omega)H(j\omega)$ and $T_o(j\omega) = \frac{G_o(j\omega)}{1 + G_o(j\omega)}$



Stability in Frequency Domain -Examples



Stability Regions or Point of Instability

Methods of determining point of instability

- Three methods of determining the crossing point into the right half plane (i.e. points of instability).
- Use Routh-Hurwitz to determine the value of K for which the system is unstable and use the magnitude condition to determine the position.
- Use the angle criterion of root locus directly and determine the crossing point. Then use the magnitude condition to determine the value of gain, K , for which the system becomes unstable.
- By substituting $s = j\omega$ in the transfer function and then using the Nyquist stability criterion to determine the frequency at which instability occurs. This frequency is the value of ω .

Example 1a

- Using Routh-Hurwitz criterion, find the value of K for which the system below is stable and determine the point of instability.

$$G(s) = \frac{K}{s(s+1)(s+2)}$$

$$F(s) = s^3 + 3s^2 + 2s + K = 0$$

Route table:	$s^3 $	1	2
	$s^2 $	3	K
	$s^1 $	$-1/3(K-6)$	0
	$s^0 $	K	0

Example 1a

- Using Routh-Hurwitz criterion, find the value of K for which the system below is stable and determine the point of instability.

$$G(s) = \frac{K}{s(s+1)(s+2)}$$

- Show that $K=6$, at the point of instability
- Determine the point of instability by using:

$$K = \frac{\prod_1^n \text{distance to open loop poles}}{\prod_1^m \text{distance to open loop zeros}}$$

$$K = x \sqrt{x^2 + 1} \sqrt{x^2 + 4} = 6$$

$$x^2 = 2$$

Example 1b

- Using the root locus angle criterion, find the value of K for which the system below is stable and determine the point of instability.

$$\sum \text{angles from zeros} - \sum \text{angles from poles} = 180^\circ \pm k360^\circ$$

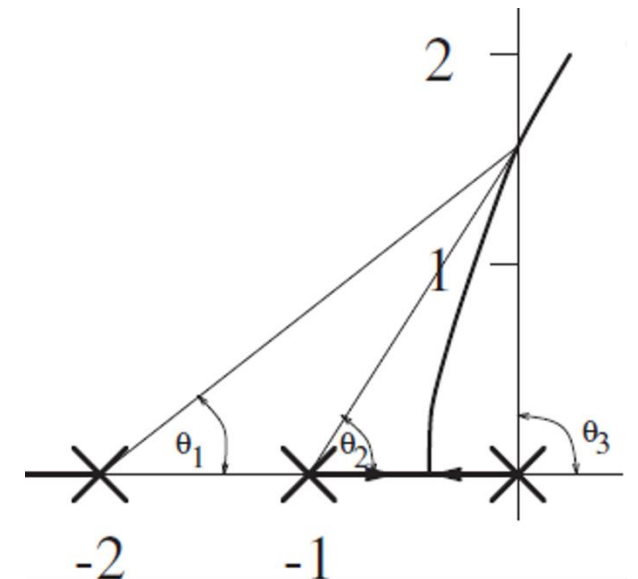
$$-\theta_1 - \theta_2 - \theta_3 = -\tan^{-1}x - \tan^{-1}\left(\frac{x}{2}\right) - 90^\circ = -180^\circ$$

$$\tan^{-1}x + \tan^{-1}\left(\frac{x}{2}\right) = 90^\circ$$

$$\tan^{-1}A + \tan^{-1}B = \tan^{-1}\left\{\frac{A+B}{1-AB}\right\}$$

$$\tan^{-1}\left\{\frac{3x/2}{1-x^2/2}\right\} = 90^\circ$$

$$1 - x^2/2 = 0 \Rightarrow x = \pm\sqrt{2}$$



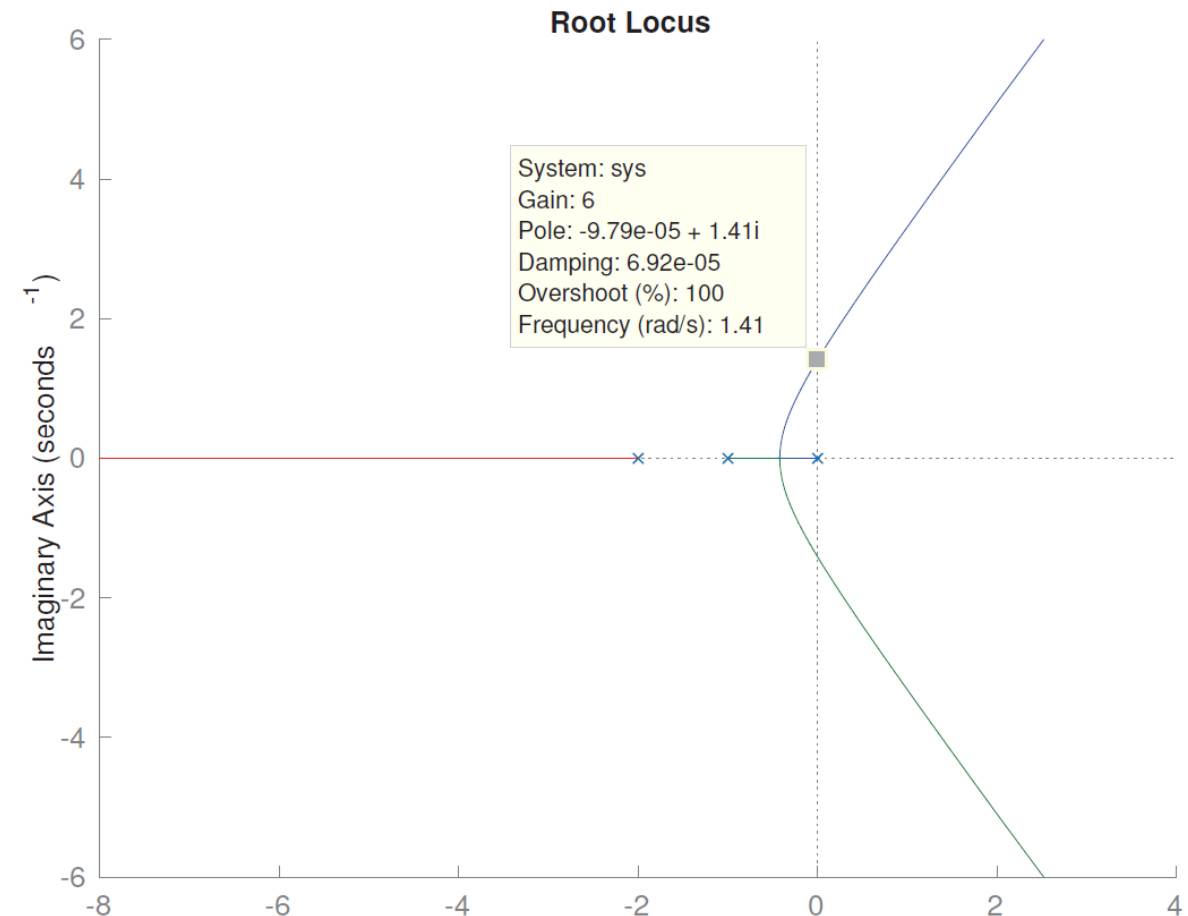
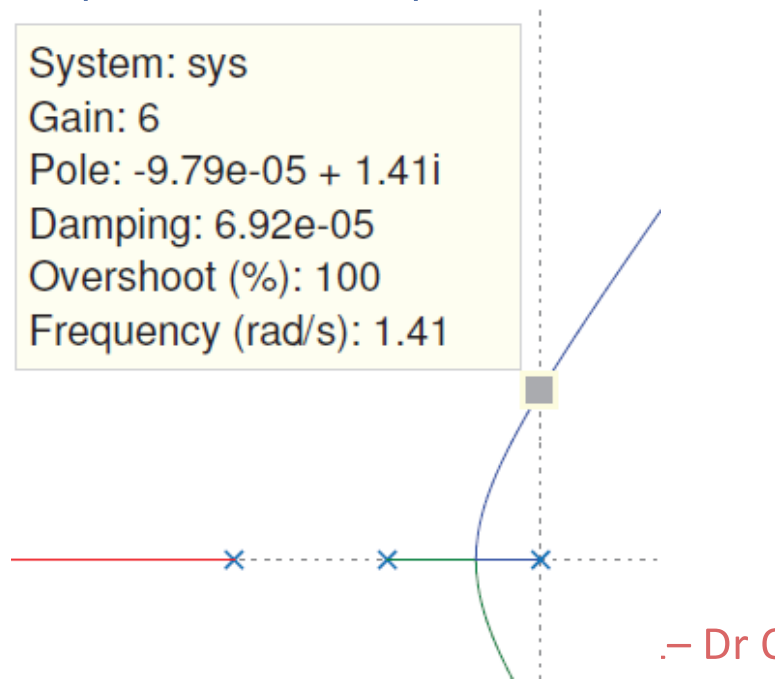
Example 1b

- Using the root locus angle criterion, find the value of K for which the system below is stable and determine the point of instability.

Recall, for points on the locus,

- $K = \frac{\prod_1^n \text{distance to open loop poles}}{\prod_1^m \text{distance to open loop zeros}}$
- $K = \sqrt{2} \times \sqrt{2+1} \times \sqrt{2+4} = 6$

System: sys
Gain: 6
Pole: $-9.79\text{e-}05 + 1.41i$
Damping: $6.92\text{e-}05$
Overshoot (%): 100
Frequency (rad/s): 1.41



Example 1c

- Using Nyquist criterion, find the value of K for which the system below is stable and determine the point of instability.

$$G(s) = \frac{K}{s(s+1)(s+2)}$$

- Two techniques can be used to determine the crossing point of the negative real axis, which both give the same solution:
 - Find the frequency at which $\text{Im}\{G(j\omega)\} = 0$
 - Find the frequency at which the phase $\arg\{G(j\omega)\} = -180^\circ$

After the frequency is determined, calculate the magnitude .

Example 1c

- Using Nyquist criterion, with the magnitude

$$|G(j\omega)| = \frac{K/2}{\omega\sqrt{1+\omega^2}\sqrt{1+\omega^2/4}}$$

$$\arg G(j\omega) = -90^\circ - \tan^{-1} \omega - \tan^{-1} \omega/2 = -90^\circ - \tan^{-1} \frac{3\omega/2}{1 - \omega^2/2}$$

$$G(j\omega) = \frac{K}{j\omega(j\omega + 1)(j\omega + 2)}; \operatorname{img}(G(j\omega)) = 0;$$

$$\frac{K/2(-j\omega + \frac{j\omega^3}{2})}{\omega^2(\omega^2 + 1)\left(\frac{\omega^2}{4} + 1\right)} = 0$$

equating this to zero implies

$$-\omega + \omega^3/2 = 0 \Rightarrow \omega = 0 \text{ or } \omega = \sqrt{2}$$

Example 1c

- Using Nyquist criterion, with the phase

Using $\arg G(j\omega) = -180^\circ$ gives

$$\tan^{-1} \frac{3\omega/2}{1 - \omega^2/2} = -90^\circ$$

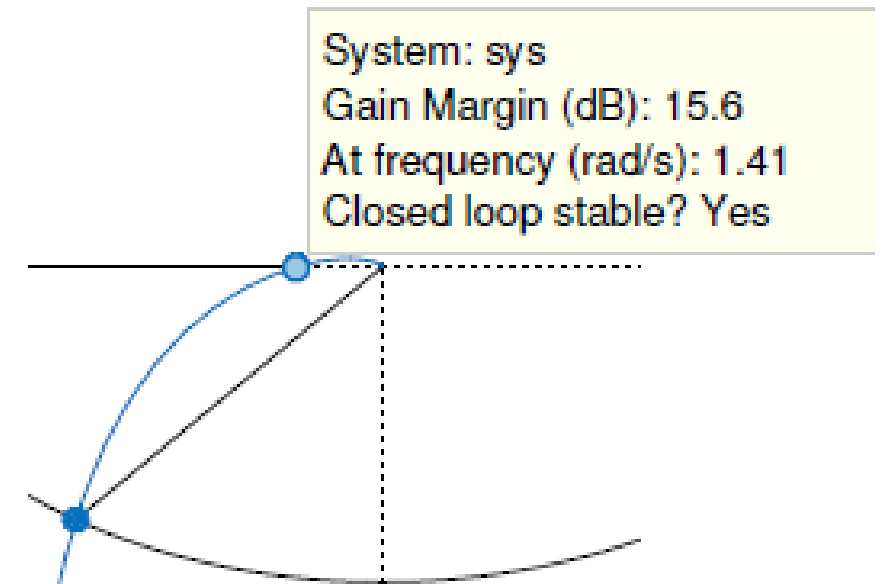
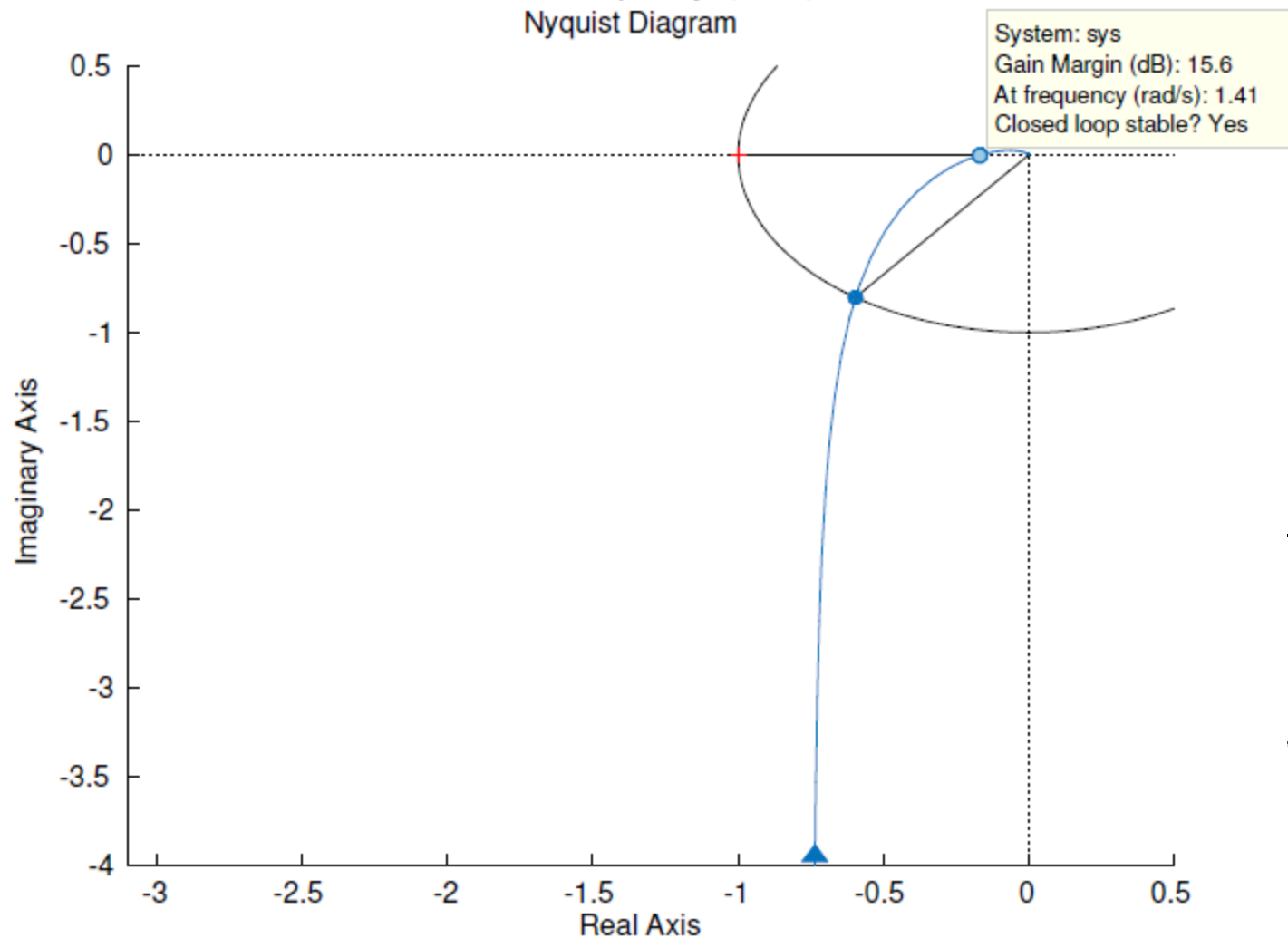
which implies that

$$\frac{3\omega/2}{1 - \omega^2/2} = \infty$$

and thus $1 - \omega^2/2 = 0$, or $\omega = \pm\sqrt{2}$. The crossing point of the locus therefore follows as above.

Example 1c

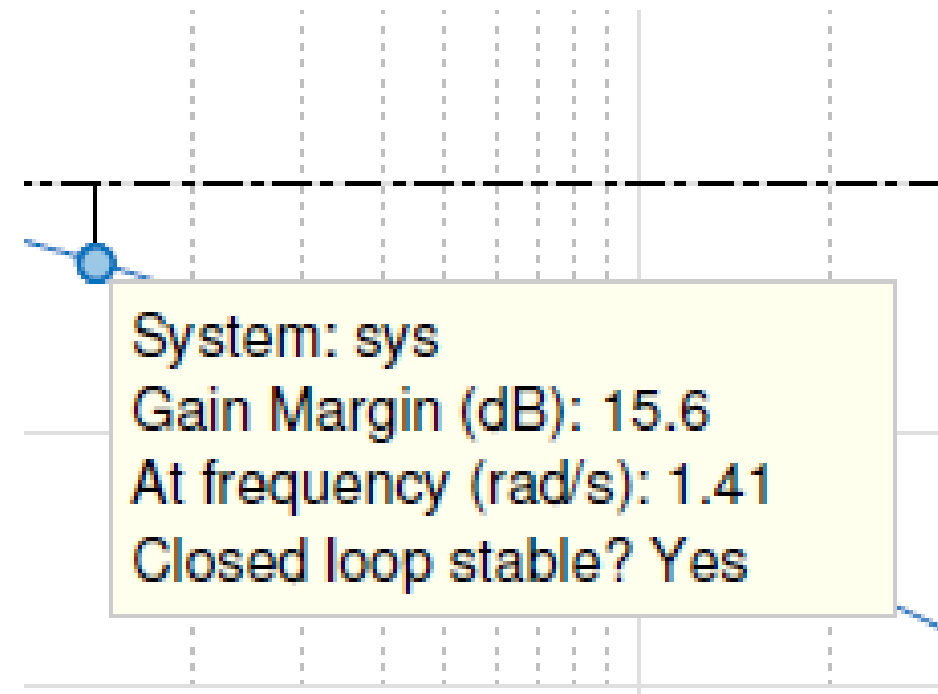
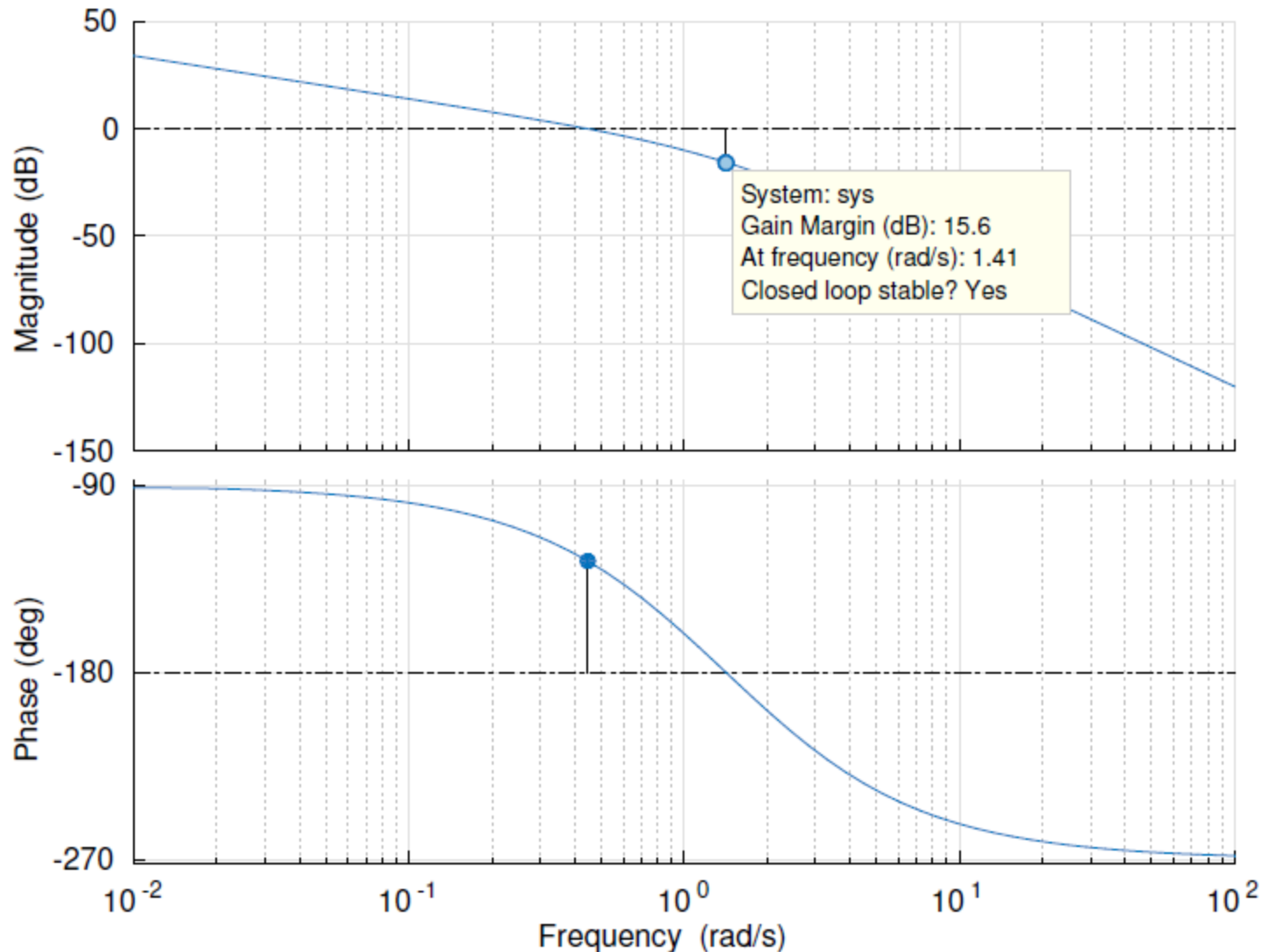
- Nyquist plot of frequency response:



Example 1c

- Bode plot of frequency response:

Bode Diagram



Example 2

- For an open-transfer function below, with $K=2$, determine if the closed-loop transfer function is stable and find the critical value for K

$$G(s)H(s) = \frac{K}{s(s+1)(2s+1)}$$

$$G(j\omega)H(j\omega) = \frac{K}{j\omega(j\omega+1)(2j\omega+1)} = \frac{K}{-3\omega^2 + j\omega(1-2\omega^2)}$$

For stability, the Nyquist plot should not encircle the $-1+j0$ point, by setting imaginary part of $G(j\omega)H(j\omega)$ as 0, $\omega = \pm \frac{1}{\sqrt{2}}$,

by substitution, $G(j\omega)H(j\omega) = -\frac{2K}{3}$, by equating this to -1 , critical value is 1.5 and stable range is $0 < K < 1.5$

Example 3

- Consider a unity feedback system with open loop transfer function below, for marginal stability find the radian frequency of oscillation

$$G(s) = \frac{K}{(s^2 + 2s + 2)(s + 2)}$$

- Show that:

$$G(j\omega)H(j\omega) = \frac{4(1 - \omega^2) - j\omega(6 - \omega^2)}{16(1 - \omega^2)^2 + \omega^2(6 - \omega^2)^2}$$

Set imaginary part to 0, find ω

$\omega = \sqrt{6}$ and show the real part yields $\left(\frac{1}{20}\right) \angle 180^\circ$.

This closed loop system is stable if the magnitude of the frequency response is less than 1 at 180 deg. Hence $K < 20$ is stable, $K > 20$ is unstable, and $K = 20$ is marginally stable. Frequency of oscillation is $\sqrt{6}$

Example 4

- Consider a unity feedback system with transfer function:

$$G(s) = \frac{K e^{-0.8s}}{s + 1}$$

Using Nyquist plot, determine the critical value of K for stability

- For this system,

$$\begin{aligned} G(j\omega) &= \frac{K e^{-j0.8\omega}}{j\omega + 1} = \frac{(K \cos(0.8\omega) - jK \sin(0.8\omega))(1 - j\omega)}{1 + \omega^2} \\ &= \frac{K}{1 + \omega^2} [\cos(0.8\omega) - \omega \sin(0.8\omega) - j(\sin(0.8\omega) + \omega \cos(0.8\omega))] \end{aligned}$$

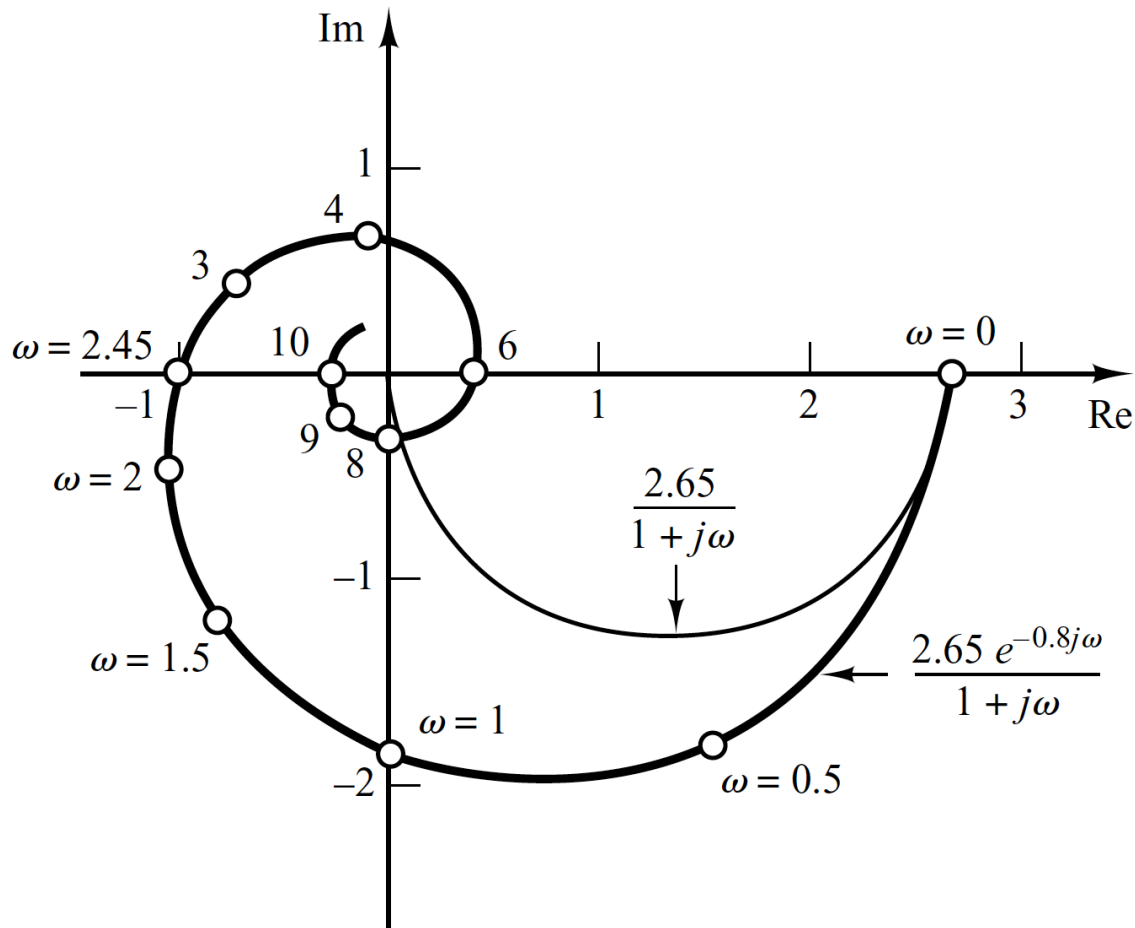
When imaginary part is zero, show $\omega = 2.4482$

Substitute this into $G(j\omega)$ and show that $K = 2.65$

Example 3

Consider a unity feedback system with transfer function:

$$G(s) = \frac{K e^{-0.8s}}{s + 1}$$

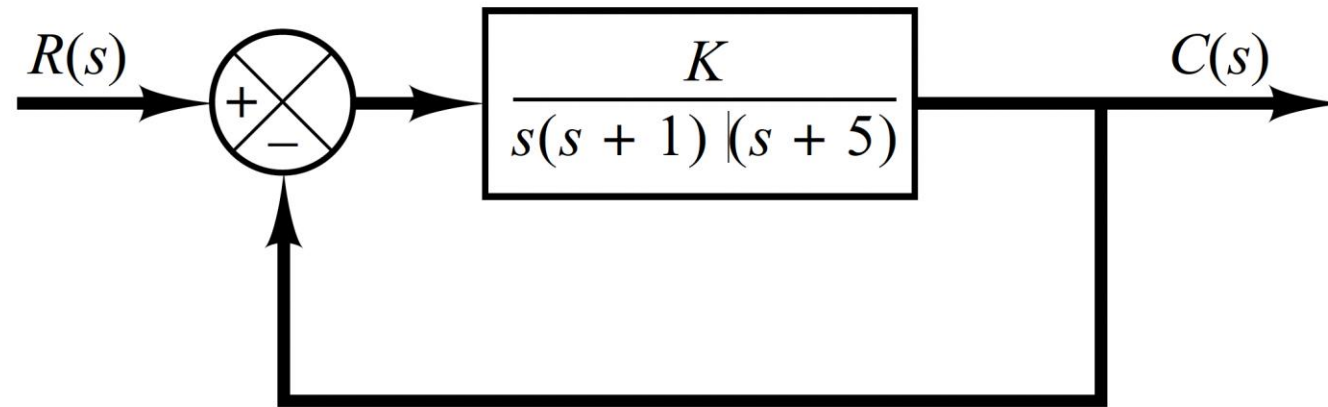




Stability Margins

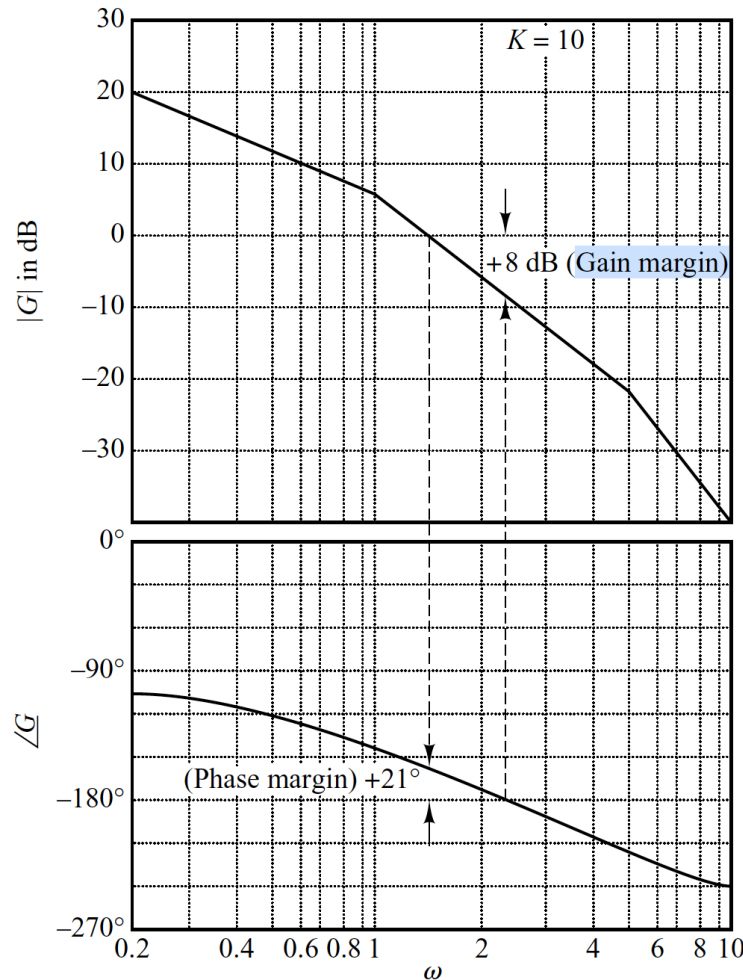
Example 1

- By drawing the bode plots, obtain the phase and the gain margins for the system below for the two cases where $K=10$ and $K=100$.

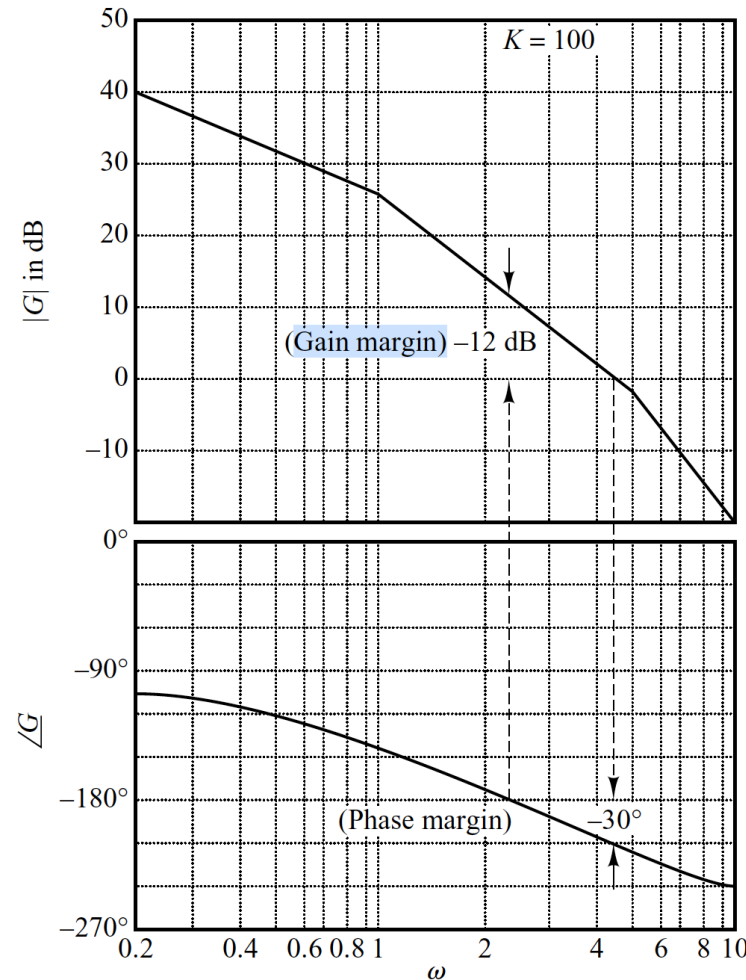


Example 1

- By drawing the bode plots, obtain the phase and the gain margins for the system below for the two cases where $K=10$ and $K=100$.



(a)

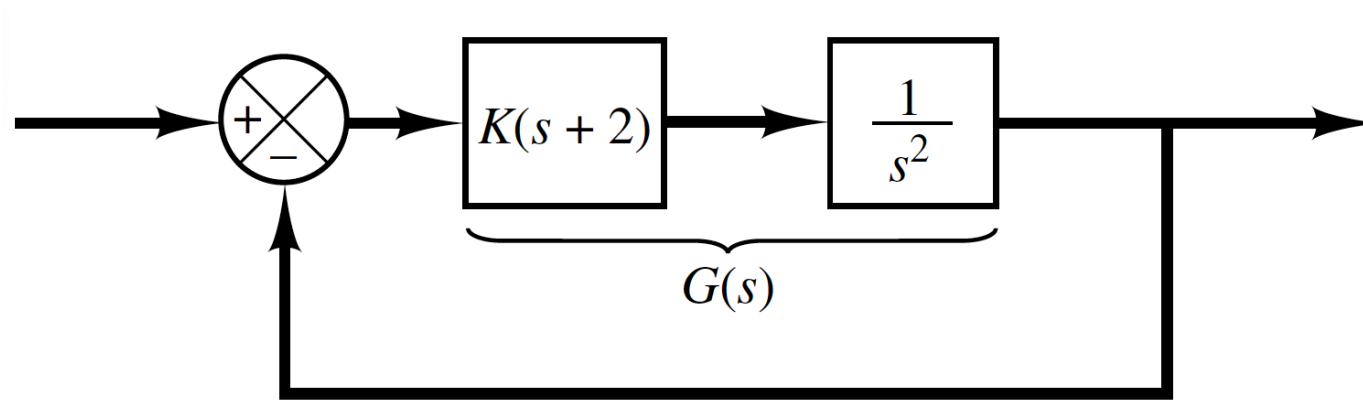


(b)

- At $K=10$
 - $GM = 8 \text{ dB}$
 - $PM = 21 \text{ deg}$
 - Stable
- At $K=100$
 - $GM = -12 \text{ dB}$
 - $PM = -30 \text{ deg}$
 - Unstable

Example 2

For the space-vehicle system below, draw the bode plot and determine the value of K so that the phase margin is 50 deg. What is the gain margin with this value of gain K ?



$$G(j\omega) = \frac{K(j\omega + 2)}{j\omega^2}$$

$$\angle G(j\omega) = \angle(j\omega + 2) - 2\angle(j\omega) = \tan^{-1}\left(\frac{\omega}{2}\right) - 180^\circ$$

For gain margin to be 50 deg, this means $\angle G(j\omega_c) = -130^\circ$ where ω_c is gain cross over frequency

Example 2

For the space-vehicle system below, draw the bode plot and determine the value of K so that the phase margin is 50 deg. What is the gain margin with this value of gain K ?

$$\tan^{-1} \left(\frac{\omega_c}{2} \right) = 50^\circ$$

$$\omega_c = 2.3835 \text{ rad/sec}$$

Since the phase curve does not cross -180 deg line, GM is infinite, the magnitude of $G(j\omega)$ must be equal to 0db at $\omega = 2.3835$

$$\frac{K(j\omega + 2)}{j\omega^2} = 1 \text{ at } \omega = 2.3835$$

$$K = 1.8259$$

Example 3

- For a standard second order transfer function, write the bandwidth as a function of damping factor and natural frequency.

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

The bandwidth ω_b (3dB point) is determined by

$$\left| \frac{C(j\omega_b)}{R(j\omega_b)} \right| = -3dB = 0.707$$

$$\left| \frac{C(j\omega_b)}{R(j\omega_b)} \right| = \frac{\omega_n^2}{(j\omega_b)^2 + 2\zeta\omega_n(j\omega_b) + \omega_n^2} = 0.707$$

$$\omega_n^4 = 0.5 \left[(\omega_n^2 - \omega_b^2)^2 + 4\zeta^2 \omega_n^2 \omega_b^2 \right]$$

Example 3

$$\omega_n^4 = 0.5 \left[(\omega_n^2 - \omega_b^2)^2 + 4\zeta^2 \omega_n^2 \omega_b^2 \right]$$

Dividing by ω_n^4 , and solving for $\left(\frac{\omega_b}{\omega_n}\right)^2$, yields

$$\left(\frac{\omega_b}{\omega_n}\right)^2 = -2\zeta^2 + 1 \pm \sqrt{4\zeta^4 - 4\zeta^2 + 2}$$

Since $\left(\frac{\omega_b}{\omega_n}\right)^2 > 0$

$$\omega_b^2 = \omega_n^2 \left(-2\zeta^2 + 1 + \sqrt{4\zeta^4 - 4\zeta^2 + 2} \right)$$

$$\omega_b = \omega_n \left(-2\zeta^2 + 1 + \sqrt{4\zeta^4 - 4\zeta^2 + 2} \right)^{\frac{1}{2}}$$

Exercise 1

For a unity feedback system with transfer function below, plot a Nyquist diagram and use your Nyquist diagram to find the range of gain K for stability.

$$G(s) = \frac{K}{(s + 2)(s + 4)(s + 6)}$$

Ans: $480 > K > -48$

Exercise 2

Find the gain margin and phase margin of the system if $K=6$

$$G(s) = \frac{K}{(s + 2)(s + 4)(s + 6)}$$

$$G_m = 38.1 \text{ dB}$$

$$P_m = \text{inf}$$



Exercise 3

For a unity feedback system with transfer function below,

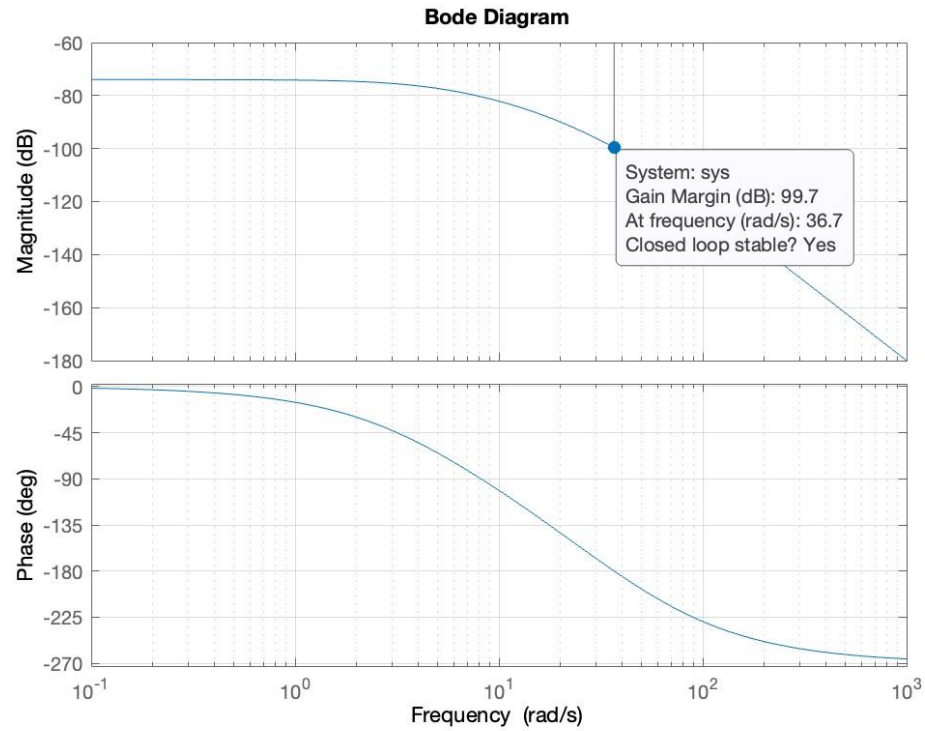
Draw Bode magnitude and phase plots

Find the range of K for stability from your plot

Evaluate the gain margin, phase margin, 0 dB frequency, and 180 deg frequency from your Bode plot for K=10,000

$$G(s) = \frac{K}{(s + 5)(s + 20)(s + 50)}$$

Exercise 3



$$96250 > K > -5000$$

$$G_m = 99.7 \text{ dB}$$

$$P_m = \infty$$

$$W_{cg} = 36.7$$

$$W_{cp} = \text{does not exist}$$